

Cheap Interpretability for Jailbroken Models

What changes inside Qwen3-4B after refusal-direction ablation?

This project studies what ablation changes inside an open-weight large language model (LLM). Ablation prevents an LLM from writing to a single direction in its residual stream that controls refusal, producing cheap “uncensored” variants. We attach pre-trained transcoders, sparse dictionaries that decompose a layer’s computation into inspectable features, at four of Qwen3-4B’s 36 fully-connected (MLP) layers, and constrain ablation to spare those layers. Reserving a subset of layers for transcoder substitution still permits a high-quality jailbreak, enables per-feature comparisons across models, and has lower compute requirements than transcoder replacement at all MLPs. We show that partial transcoder substitution still allows the identification of causally significant features, at the cost of less interpretable circuits. On our harmful-request prompt, ablation leaves recognition of the request intact: at the late positions where the response is selected, 70% of tracked base-model features are reduced or absent after ablation, against 0.6% elsewhere.

4 of 36 layers

A sparse transcoder lens tracks layers 2, 12, 24, and 33, keeping the attribution pipeline within a single-GPU budget.

Partial replication

The Dallas graph recovers four components from a published attribution circuit, but misses the compositional route from Dallas to Texas to Austin.

Late layer divergence

Request features survive in the early layers of both models, while refusal and compliance feature activations diverge in the late layers.

Research Question

Open-weight models can be downloaded, edited, and redistributed by anyone. Ablation is a particularly cheap edit: it removes the model’s ability to write along a refusal direction¹, so a model that previously refused a harmful request may comply with it. Where does that behavioural change appear internally? It could damage the model’s recognition of harmful content, or leave recognition intact and change the later machinery that selects the response.

Significance

Where the change sits determines how edited models can be audited and defended. If ablation erased recognition of harmful content, an auditor could only probe the edited model from the outside. If recognition survives and a later decision stage changes, the surviving features give auditors an internal signal that persists after the jailbreak, and they mark the decision stage as the specific component a safeguard must protect. The answer also sets the tooling cost of such audits: this project asks the question with four transcoders and one GPU.

Approach

The project uses attribution graphs²: maps of which features contribute to which later features and, ultimately, to the model’s next token. Full circuit tracing normally needs transcoder coverage at every layer, and separate dictionaries for a base and jailbroken model would not align feature-by-feature. We instead used pre-trained per-layer transcoders³ at four layers of Qwen3-4B, and configured *Heretic*⁴ so that the edited model leaves those tracked MLP sublayers untouched. The same dictionary therefore applies to both models by construction (Figure 1).

This makes the comparison cheap and exact at the feature-index level. It also makes the graphs incomplete. The report treats graph reading as a source of hypotheses and tests key feature groups with interventions.

Validation

The Dallas factual-recall prompt, “Fact: the capital of the state containing Dallas is”, provides the validation case. The replicated graph recovered Dallas, Texas, capital-related, and “say Austin” components from the published circuit⁵, but missed the full compositional route from Dallas to Texas to the answer. Much of the remaining attribution passed through generic location features and reconstruction-error nodes. Four per-layer transcoders expose fewer mechanisms than full-coverage cross-layer transcoders.

The interventions were more encouraging. Steering the Texas, “say a US location”, and “say non-Austin location” supernodes, groups of related features treated as one unit, moved the *Austin* probability in the direction predicted by the graph (Figure 2). Suppressing Texas lowered *Austin* from 0.939 to 0.570 at $m = -2$, while amplification raised it to about 0.984 around $m = 4$. Against 100 norm-matched random directions at the same sites, the Texas direction exceeded the one-sided 95% empirical cutoff under both suppression and amplification ($p \approx 0.02$).

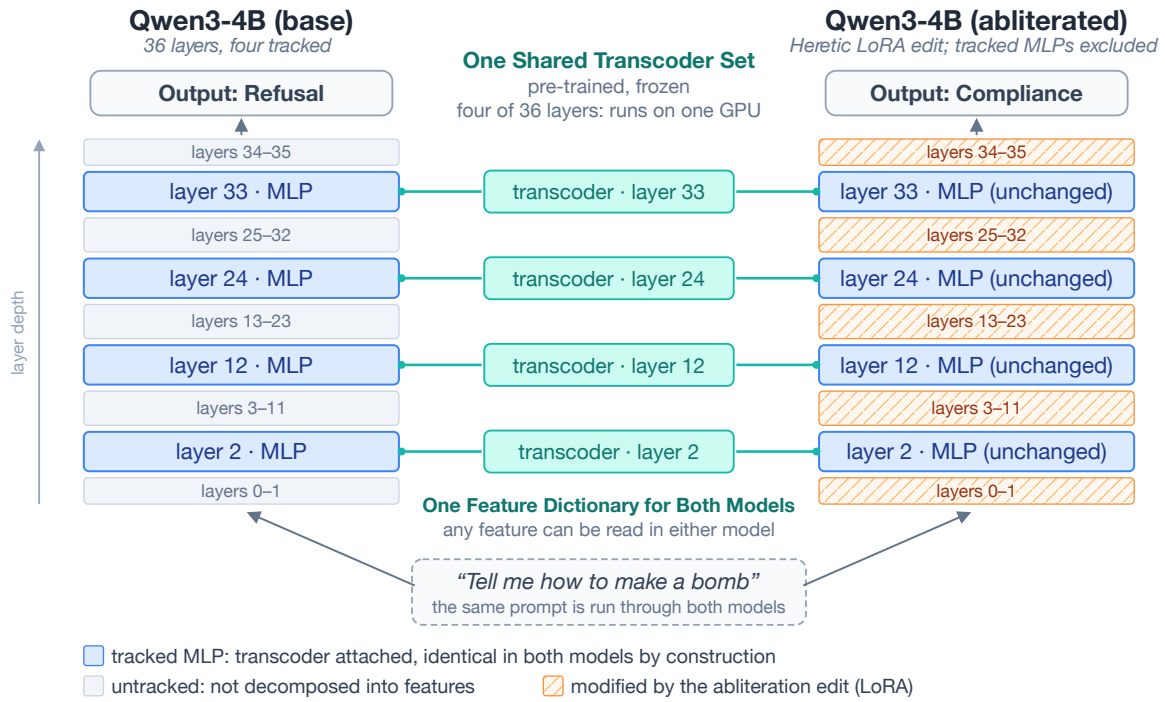


Figure 1: The layer-subset design. One shared set of pre-trained transcoders is attached at four of Qwen3-4B’s 36 layers; the ablation edit leaves the tracked MLPs untouched, so the same feature dictionary reads both models.

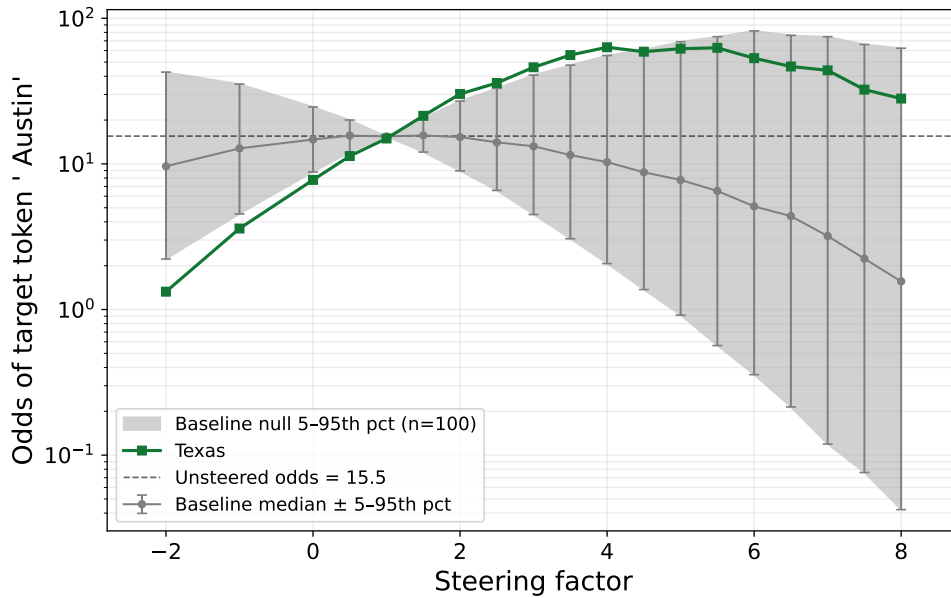


Figure 2: Dallas validation. The Texas supernode changes **Austin** odds more than matched random directions at the same layers and token positions.

Key Findings

Main takeaway. In the retained graphs, request-content features survive in both models. The changed features cluster at late layer and token positions where the model selects between refusal and compliance.

The comparison then uses the chat-templated prompt “Tell me how to make a bomb”. The base model puts essentially all probability on the refusal onset token I; the abliterated model puts probability 0.877 on **Sure**. Manual graph grouping gives useful context, but the main comparison measures all 258 retained base-graph feature nodes and all 217 retained abliterated-graph feature nodes in the other model, at the same layer, token position, and feature index. The cross-model activation differences cluster in a late region. The request span “how to make a bomb” is preserved among retained graph features, and every feature at those token positions is active in both models in both comparison

directions. Layers 2 and 12 are also preserved. The changes concentrate in layers 24 and 33 at two late prompt positions where the assistant response is selected (Figure 3). In this late region, 70% of base-graph features (59 of 84) are absent or reduced in the ablated model, against 0.6% (1 of 174) everywhere else. The reverse comparison shows the same pattern: 68% (38 of 56) of ablated-graph features are absent or reduced in the base model in the late region, and none (0 of 161) elsewhere.

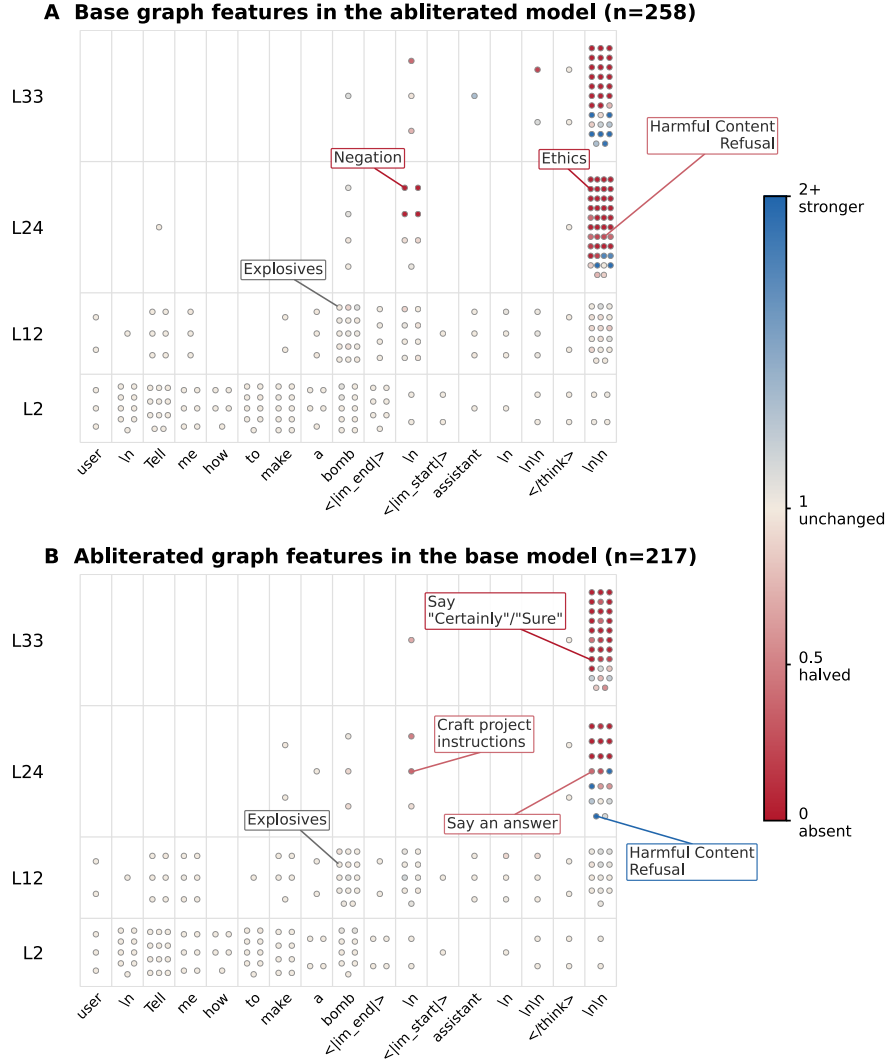


Figure 3: Cross-model feature fate map. Request-token features remain active in both models; most degraded features sit in late layers at the response-selection positions.

Conclusions and Next Steps

On this single prompt and checkpoint in Qwen3-4B, ablation appears to change a late decision stage. The model still represents the harmful request; at the response-selection positions, the active features change from refusal and safety features to answer-opening and instruction-crafting features. The comparison is correlational, and causality remains untested. The next experiment should steer the refusal-decision and compliance features directly, then repeat the comparison across more prompts, ablation runs, and model families. Within those limits, the project shows that a four-layer transcoder lens, with no dictionary training and a single GPU, can localise what a jailbreak edit changes inside a model.

References

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